University of Central Punjab
Faculty of Information Technology and Computer Science

Course Title: Information Security
Course Code: CSNC 3413

Assignment 2

Registeration#	Name

CLO	CLO STATEMENT	Bloom's Taxonomy Level	PLO
1	Identify and comprehend basics terms and technologies and concepts of information security	2	1
2	Investigate and analyze real situations from the information security point of view and model them using various security control measures	4	3

Instructions:

1. **Attempt all questions.**
2. **Write your answer showing all steps required to perform the task.**
3. **Assignment must be solved on A4 sheets or Assignment sheets only. Violation will result to deduction of 1 mark from the scored marks. Submissions will be made in soft form via portal and hard form in the class.**
4. **Each student will have attached this front page with his/her assignment. Violation will result to deduction of 1 mark from the scored marks.**
5. **Due Date for Assignment is: 13-11-25.**
6. **Late submission will result in 10% deduction in marks.**
7. **No request for late submissions will be considered after two working days of the deadline.**
8. **No request to review assignment will be considered after 3 working days of the review in class.**

Problem 01: Complete DES Rounds. (10 marks)

Given the following inputs for one round of DES:

- $L_0 = 12345678_{16}$
- $R_0 = 23456789_{16}$
- Subkey $K_1 = AABB09182736CCDD_{16}$

Question: Perform the following steps for one DES round:

1. Expand R_0 to 48 bits using the Expansion function.
2. XOR the expanded R_0 with K_1.
3. Pass the result through the 8 S-Boxes.
4. Apply the P-Box permutation.
5. XOR the result with L_0.

What are L_1 and R_1 after the round?

Problem 02: RC4 (10 marks)

Solve questions of RC4 with following considerations:

1) Length of State Array = 5

2) Length of key is 4 Bytes.

Message (M)

Byte 1: 01000001

Byte 2: 01101110

Key:

K[0] = 00001111

K[1] = 10001011

K[2] = 00101101

K[3] = 00100000